

Analysis VI

Problem Set #1

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Ilyas Zanan

Problem 1. Let F be defined as:

$$F(x) = \int_0^1 \frac{dt}{(t^2 + 1)(t^2 + x^2)}$$

1. Show that F is continuous on $(0, +\infty)$.

2. Deduce the value of the integral:

$$\int_0^1 \frac{dt}{(t^2 + 1)^2}$$

Solution.

1. We have

$$f(x, t) = \frac{1}{(x^2 + t^2)(t^2 + 1)}$$

$\forall x \in \mathbb{R}_*^+$ and $t \in [0; 1]$ $f(x, t)$ is a continuous function then it is integrable. we have

$$|f(x, t)| = \left| \frac{1}{(t^2 + 1)(t^2 + x^2)} \right| \leq \frac{1}{(t^2 + 1)(t^2 + a^2)} = g(t)$$

and g is clearly integrable then F is continuous in $(0, \infty)$.

2. we have

$$\begin{aligned} f(x, t) &= \frac{1}{x^2 - 1} \left(\frac{1}{t^2 + 1} - \frac{1}{t^2 + x^2} \right) \\ F(x) &= \frac{1}{x^2 - 1} \left(\int_0^1 \frac{dt}{t^2 + 1} - \int_0^1 \frac{dt}{t^2 + x^2} \right) \\ &= \frac{1}{x^2 - 1} \left(\frac{\pi}{4} - \frac{1}{x} \arctan \frac{1}{x} \right) \end{aligned}$$

now for $x = 1$ we get

$$\int_0^1 \frac{dt}{(t^2 + 1)^2} = \lim_{x \rightarrow 1} F(x) = \frac{\pi}{8} + \frac{1}{4}$$

Problem 2. Let $x > 0$ and $b \neq 0$. We define:

$$F(x) = \int_0^{+\infty} e^{-tx} \sin(bt) \frac{dt}{t}$$

1. Show that F is differentiable on $(0, +\infty)$ and calculate $F'(x)$.

2. Calculate $\lim_{x \rightarrow +\infty} F(x)$ and $\lim_{x \rightarrow 0} F(x)$.

3. Deduce the value of the integral:

$$\int_0^{+\infty} \frac{\sin(bt)}{t} dt$$

Solution.

1. we have

$$f(x, t) = \frac{e^{-tx} S_{bt}}{t}$$

$$\frac{\partial f}{\partial x} = e^{-tx} S_{bt}$$

$$\forall x > a > 0$$

$$\left| \frac{\partial f}{\partial x} \right| = |e^{-tx} S_{bt}| \leq e^{-ta}$$

$\int_0^\infty e^{-at} dt$ converges for $a > 0$.

$$F'(x) = \int_0^\infty -e^{-tx} S_{bt} dt$$

$$= \frac{-b}{x^2 + b^2}$$

2. (a) $x \rightarrow \infty$:

$$\lim_{\infty} F(x) = \int_0^\infty \lim_{\infty} e^{-tx} \frac{S_{bt}}{t} dt$$

(b) $x \rightarrow 0$:

$$\lim_0 = F(0) = \int_0^\infty \frac{S_{bt}}{t} dt$$

3.

$$F(x) = \int \frac{-b}{x^2 + b^2} dx = -\arctan \frac{x}{b} + C$$

$$0 = \lim_{\infty} (-\arctan \frac{x}{b} + C)$$

$$\begin{cases} \text{if } b > 0 \implies C = \frac{\pi}{2} \\ \text{if } b < 0 \implies C = -\frac{\pi}{2} \end{cases}$$

$$F(0) = \int_0^\infty \frac{S_{bt}}{t} dt = \frac{\pi}{2} (-1)^n$$

Problem 3. Consider the functions f and g defined on $\mathbb{R}^+$ by:

$$f(x) = \int_0^\infty \frac{e^{-xt}}{1+t^2} dt \quad \text{and} \quad g(x) = \int_0^\infty \frac{\sin t}{t+x} dt$$

1. Show that f and g are of class C^2 on $\mathbb{R}^{+*}$ and that they satisfy the differential equation:

$$y'' + y = \frac{1}{x}$$

2. Show that f and g are continuous at 0.

3. Show that $\lim_{x \rightarrow +\infty} f(x) = \lim_{x \rightarrow +\infty} g(x) = 0$.
4. Deduce that $g(x) = f(x)$, then give the value of the integral:

$$\int_0^\infty \frac{\sin t}{t} dt$$

Solution.

1. (a)

$$\begin{aligned} f'(x) &= \int_0^\infty \frac{e^{-xt}}{1+t^2} dt \\ f''(x) &= \int_0^\infty \frac{t^2 e^{-xt}}{1+t^2} dt \\ f''(x) + f(x) &= \frac{1}{x} \end{aligned}$$

(b)

$$\begin{aligned} g'(x) &= \int_0^\infty \frac{S_t}{t+x} dt \\ g''(x) &= \int_0^\infty \frac{2S_t}{(t+x)^3} dt \\ g''(x) + g(x) &= \frac{1}{x} \end{aligned}$$

2. (a) $f(0) = \int_0^\infty \frac{1}{1+t^2} dt = \arctan(t)|_0^\infty = \frac{\pi}{2}$; it converges uniformly to 0.
 (b) $g(0) = \int_0^\infty \frac{S_t}{t} dt$ converges conditionally, g is continuous at 0.
3. $\lim_{\infty} f(x) = 0$; $|g(x)| \leq \int_0^\infty \frac{|S_t|}{t+x} dt \implies \lim_{\infty} g(x) = 0$.
4. (a) both satisfy the differential equation, their difference $h(x) = f(x) - g(x)$ satisfies $h'' - h = 0$ meaning $h(x) = AC_x + BS_x$.
 (b) since $\lim_{\infty} f(x) = \lim_{\infty} g(x) = 0 \implies A = B = 0 \implies f(x) = g(x)$
 (c) since $f(0) = g(0)$

$$\int_0^\infty \frac{S_t}{t} dt = \frac{\pi}{2}$$

Problem 4. Let x, y be two strictly positive real numbers. We define:

$$F(x, y) = \int_0^\infty \frac{e^{-xt} - e^{-yt}}{t} dt$$

1. For a fixed $y > 0$, show that $x \mapsto F(x, y)$ is of class C^1 on $\mathbb{R}^{+*}$.
2. Calculate $\frac{\partial F}{\partial x}(x, y)$.
3. Deduce the value of $F(x, y)$.

Solution.

Problem 5. Let a, b be two strictly positive real numbers. We define:

$$F(x) = \int_0^{\infty} \frac{e^{-at} - e^{-bt}}{t} \cos(xt) dt$$

1. Show that the domain of definition of $F(x)$ is $\mathbb{R}$.
2. Show that F is of class C^1 on $\mathbb{R}$ and calculate $F'(x)$.
3. Show that $\lim_{x \rightarrow +\infty} F(x) = 0$. Deduce the expression for $F(x)$.

Solution.

Problem 6. Consider the application $\Gamma : \mathbb{R} \rightarrow \mathbb{R}$ defined by:

$$\Gamma(x) = \int_0^{\infty} t^{x-1} e^{-t} dt$$

1. Give the domain of definition, D_{Γ} , of the "Euler Gamma" function Γ .
2. Prove the formula $\Gamma(x+1) = x\Gamma(x), \forall x \in D_{\Gamma}$. Deduce that $\Gamma(n+1) = n!$. Show that $\Gamma(x) \sim_0 \frac{1}{x}$.
3. Show that the function Γ is convex and provide its variation table.
4. Calculate $\Gamma(\frac{1}{2})$ and deduce $\Gamma(n + \frac{1}{2})$ for all $n \in \mathbb{N}^*$.
5. Let $I_n = 2 \int_0^{\infty} u^n e^{-u^2} u$. Calculate I_{2n} and I_{2n+1} .

Solution.